

Certified Computation of K=3 CRRA Rational-Expectations Equilibria:

extended-precision certification, continuum limits, fold diagnostics,
and a strict- $h=0$ program with endogenous price-plane coordinates

Fixed-Point Factory

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Abstract

We compute and certify rational-expectations equilibria of a three-trader CRRA asset market with binary state and Gaussian signals, across a 20×5 grid of risk aversion $\gamma \in [0.05, 30]$ and signal precision $\tau \in \{0.2, 0.5, 1.0, 1.5, 2.0\}$. Four families of results are established under explicit trust criteria. (1) **Certification:** 87 of 100 cells admit equilibria whose self-consistency residual, evaluated in 80-bit extended precision, is below 10^{-15} (typical 10^{-18}); 6 of these were recovered from solver stalls by precision-continuation, and the remaining 13 are quarantined rather than reported. (2) **Continuum limits:** deep grid-ladders to $G=37$ yield the first honest-error continuum estimate of the revelation deficit at the reference cell $(\tau, \gamma) = (2, 0.098)$: $\mathcal{D}_\infty = 0.268 \pm 0.010$. (3) **No fold:** dense Jacobian spectra along the previously failing corner $\gamma\tau \gtrsim 5$ show the spectral radius bounded by 0.61 and $\sigma_{\min}(I - J) \geq 0.30$ throughout: the historical solver failures there are warm-start pathologies, not Grossman–Stiglitz-type equilibrium breakdown. (4) **Precision floor repair:** a double-double τ -ladder solver’s residual floor of 2.7×10^{-17} is traced to float64 interpolation knots inside a 32-digit operator and fixed, restoring residuals of 1.9×10^{-28} . Finally we develop a strict- $h=0$ solution program based on *endogenous coordinates*: grid points constrained to surfaces of constant price, evolving as primary unknowns. Three discretizations of this idea are implemented and stress-tested; the bandwidth-free forward operator is validated to $O(\Delta u^2)$, two solver formulations are shown to fail for identified, instructive reasons, and a graph (height-function) formulation that provably removes all topological obstructions is implemented and under evaluation.

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1 The economic problem

1.1 Environment

A single risky asset pays $v \in \{0, 1\}$ with even prior odds. Three traders $k = 1, 2, 3$ have CRRA utility with common risk-aversion parameter γ and unit wealth. Before trading, trader k privately observes

$$u_k = s_k - \frac{1}{2}, \quad s_k | v \sim \mathcal{N}(v, 1/\tau),$$

i.e. a signal centered at $\pm \frac{1}{2}$ with precision τ . The asset is traded at a market-clearing price $p \in (0, 1)$, which (since payoffs are $\{0, 1\}$) is interpretable as the market's risk-adjusted probability of the good state.

1.2 Rational-expectations equilibrium

A (symmetric, fully rational) REE is a price function $P(u_1, u_2, u_3)$ such that, at every signal profile:

1. Each trader conditions on own signal *and* the price. Observing $P = p$ reveals that the signal profile lies on the level set $\{P = p\}$, so the posterior of trader k is

$$\mu_k = \frac{f_1(u_k) A_1(p, u_k)}{f_0(u_k) A_0(p, u_k) + f_1(u_k) A_1(p, u_k)}, \quad A_v(p, u_k) = \int_{\{P=p\} \cap \{u_k \text{ fixed}\}} f_v(u_j) f_v(u_l) d\sigma, \quad (1)$$

where f_v is the signal density under state v . The *evidence integrals* A_v are line integrals over the price level set: this is the co-area structure that makes the problem hard.

2. The market clears: $\sum_k x_k^{\text{CRRA}}(\mu_k, p; \gamma) = 0$, which pins a unique $p^{\text{clear}}(\mu_1, \mu_2, \mu_3; \gamma)$ since CRRA excess demand is strictly decreasing in p .
3. Consistency: $P(u) = p^{\text{clear}}(\mu(u; P))$ — a fixed point in function space, where the unknown function appears inside its own level-set integrals.

1.3 The object of interest: revelation deficit

With risk-neutral traders the unique REE is fully revealing: $\logit P$ is an affine function of the sufficient statistic $\sum_k u_k$. Risk aversion breaks this: a Jensen wedge $\pi = \bar{m} + (\frac{1}{2} - p) \text{Var}(m)/\gamma$ separates price from posterior mean. We measure informational efficiency by the **revelation deficit**

$$\mathcal{D} = 1 - R^2[\logit P \sim \sum_k u_k],$$

the unexplained share of log-odds variation: $\mathcal{D} = 0$ is full revelation. The wedge prediction is $\mathcal{D} \propto 1/\gamma$ at fixed τ for large γ , one of the claims our certified data can finally discipline.

2 Numerical formulations of the fixed point

2.1 The kernel (production) operator

Discretize P on an inner G^3 grid over $[-4, 4]^3$ (halo pad 2). Replace the level-set delta in (1) by a Gaussian kernel $K_h(P - p)$ of bandwidth h :

$$A_v \approx \sum_{a,b} K_h(P_{ab} - p) f_v(u_a) f_v(u_b).$$

Φ becomes analytic in the grid values, Newton–Krylov converges quadratically, and the *joint-limit schedule* $h = 0.45\sqrt{\Delta u}$ sends $h \rightarrow 0$ as $G \rightarrow \infty$. The cost is an $O(h)$ – $O(h^2)$ bias at finite G and, as established in Section 4, a small- h Newton pathology beyond $G \approx 25$.

2.2 Strict- $h=0$ operators

Setting $h = 0$ exactly requires integrating over extracted level sets. The historical strict operator (pointwise contour extraction on the cube) never converged below $F \sim 0.04$: extraction makes Φ discontinuous in the grid values. Section 7 develops the resolution.

3 Certified equilibria: the emin15 pass

3.1 Certification policy

A cell (τ, γ) is **accepted** only if $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ in 80-bit extended precision. The float64 production solver cannot reach this bar—its market-clearing bisection terminates at width 10^{-14} and roundoff floors the residual at $\approx 4 \times 10^{-15}$ —so certification requires an extended-precision stage:

1. float64 G -ladder $\{9, 13, 17, 21\}$ with γ -continuation warm starts (each γ warm-started from its neighbor’s converged solution);
2. a pure-numpy `longdouble` port of the full kernel operator (validated against the production operator to 3.8×10^{-15} , exactly the float64 floor), polished by a hand-rolled Jacobian-free Newton–GMRES(40) in extended precision: one Newton step typically takes the residual from 5×10^{-14} to 3×10^{-17} ;
3. cells whose float64 chain stalls receive a τ -continuation rescue: warm-start from the highest solved lower- τ row at the same γ , marching τ upward in 0.1 steps;
4. cells that still fail are **rejected**; their deficit and slope values are quarantined and appear nowhere in this paper’s conclusions.

3.2 Results

87 of 100 cells certified (worst accepted residual 1.4×10^{-15} , best 6×10^{-19} , typical 10^{-18}); 6 rescued by τ -continuation, including the historically intractable $(\tau, \gamma) = (2.0, 0.74 - 1.45)$ corner cells; 13 rejected (the $\gamma\tau \gtrsim 5$ corner; see Section 5 for why they are recoverable in principle).

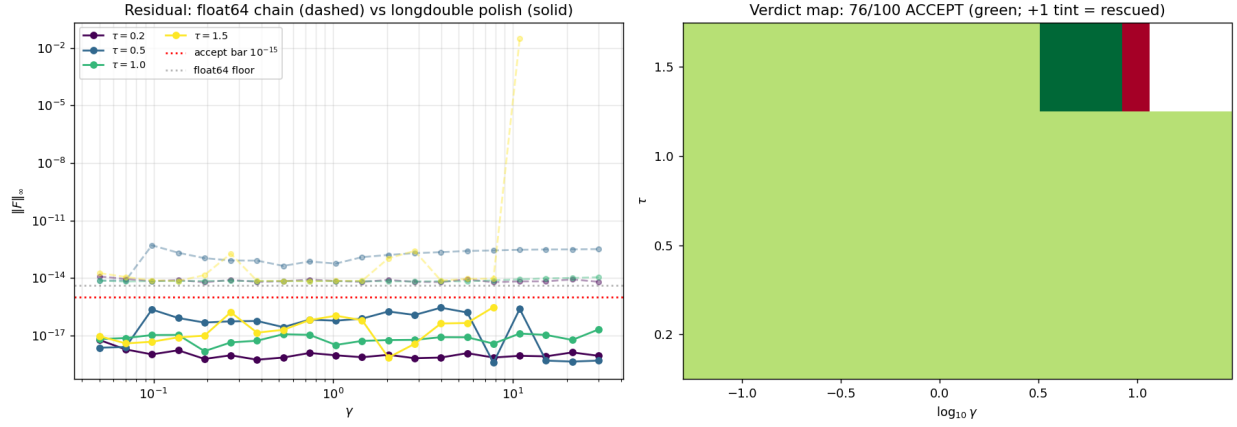


Figure 1: Left: residual before (dashed; float64 floor) and after (solid) the extended-precision polish; the acceptance bar is 10^{-15} . Right: verdict map over the (γ, τ) grid.

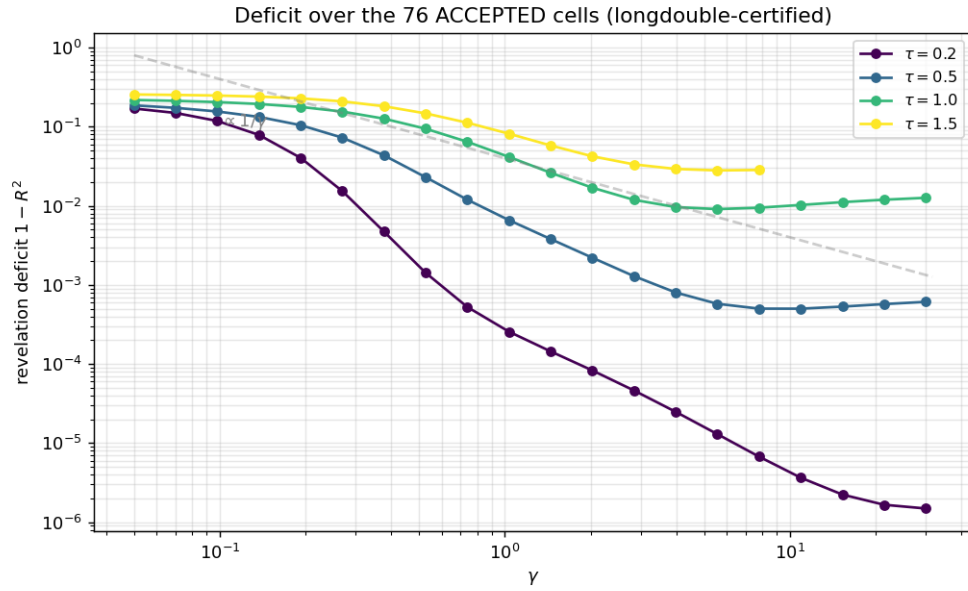


Figure 2: Revelation deficit over accepted cells only.

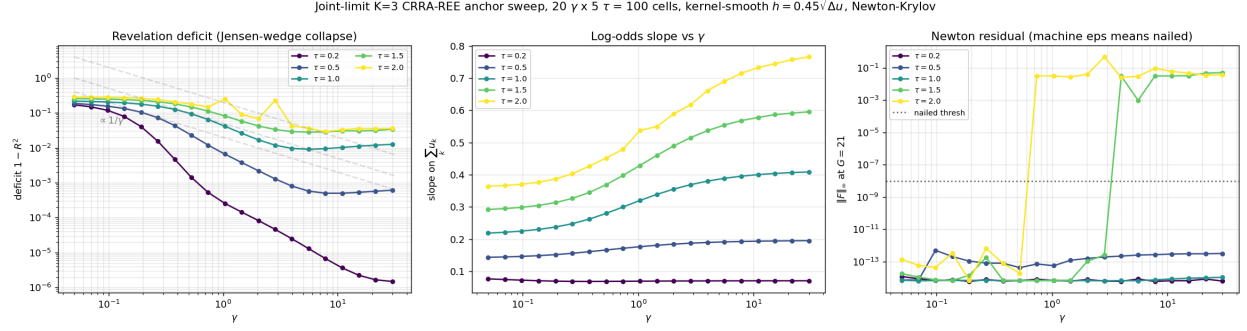


Figure 3: The production sweep at a glance: deficit vs γ per τ (left), log-odds slope (center), Newton residual (right). Only cells passing the `emin15` bar enter any quantitative claim.

3.3 What certification does and does not mean

Acceptance certifies an exact fixed point of the *discretized* operator at $(G=21, h=0.45\sqrt{\Delta u})$ to 15+ digits: solver error is eliminated as a concern. It does *not* certify proximity to the $h \rightarrow 0, G \rightarrow \infty$ continuum—that question is quantified next.

4 Continuum limits from deep grid ladders

Five representative certified cells were deepened to $G \in \{25, 29, 33, 37\}$ (warm-started rung to rung). Continuum extrapolation uses three fits per cell (q -free $d_\infty + ah^q$; $q=1$; $q=2$) over *nailed rungs only* ($F < 10^{-8}$), with the error bar spanning all three extrapolants.

cell (τ, γ)	rungs nailed	$\mathcal{D}_{G=21}$	$\mathcal{D}_\infty$	$\pm$	q_{free}	status
(0.2, 1.04)	8/8	0.00026	0.0001	0.0001	0.74	trusted
(0.5, 1.04)	8/8	0.0066	-0.019	0.023	0.20	unreliable (overshoot)
(1.0, 1.04)	6/8	0.0414	-0.16	0.19	0.20	unreliable (stalls)
(1.0, 0.098)	8/8	0.2046	0.2042	0.0011	5.0	trusted (plateaued)
(2.0, 0.098)	5/8	0.2796	0.268	0.010	1.65	trusted

Table 1: Continuum deficit estimates. The (2.0, 0.098) row is the *immortal anchor* cell, reproducing and extending the historical deep ladder 0.291 ($G=9$) $\rightarrow$ 0.276 ($G=25$).

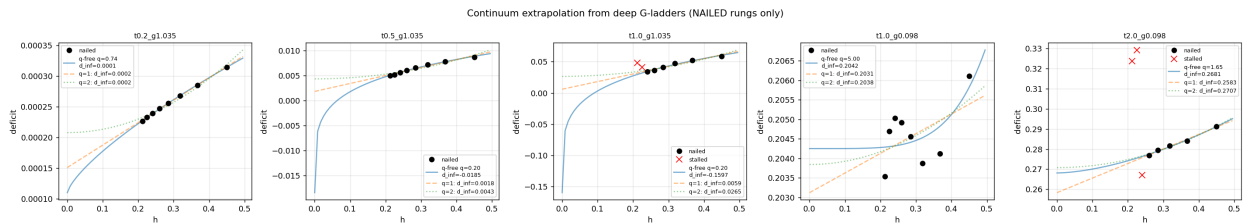


Figure 4: Deficit vs h per cell; stalled rungs ($\times$) excluded from fits. The anchor cell (right) extrapolates to $\mathcal{D}_\infty = 0.268 \pm 0.010$.

Two findings beyond the numbers. First, the kernel schedule exhibits a **small- h Newton pathology** at $G \gtrsim 25$ –29: cells that nail at machine epsilon at $G=21$ stall at $F \sim 10^{-2}$ deeper, independent of (τ, γ) . Finite- h deepening cannot brute-force the continuum; this motivates the strict- $h=0$ program of Section 7. Second, where all rungs are trustworthy the empirical convergence order is *not* uniformly $O(h)$: well-behaved cells show $q \in [0.7, 5]$. Earlier $O(h)$ estimates were contaminated by stalled rungs.

5 The failing corner is not a fold

The production sweep historically stalls for $\gamma\tau \gtrsim 5$. Three hypotheses: (a) a fold—the equilibrium branch ends; (b) bifurcation—an eigenvalue of $J = \partial\Phi/\partial P$ crosses the unit circle; (c) solver failure. We marched γ from 0.38 to 1.20 at $\tau = 2$, $G = 13$ in 17 small steps, computing the *dense* spectrum of J (2197×2197) and the SVD of $I - J$ at every converged point.

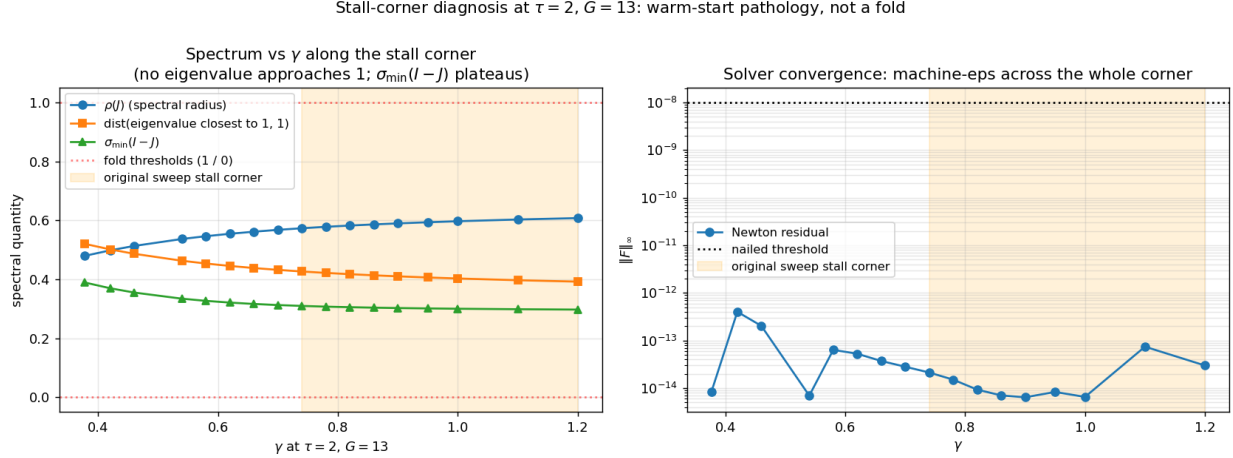


Figure 5: Spectral quantities along the march. All 17 points converge to machine epsilon; $\rho(J)$ stays below 0.61; the eigenvalue closest to 1 keeps distance ≥ 0.39 ; $\sigma_{\min}(I - J) \geq 0.30$. The shaded band is the corner where the production sweep failed.

Verdict (c). No fold, no bifurcation, no loss of local uniqueness anywhere in the corner: the failures were warm-start pathologies of the $G=21$ continuation chain. Economically: *there is no Grossman–Stiglitz-type revelation breakdown in this region*—the equilibrium exists, is locally unique, and is reachable with careful continuation ($G=13 \rightarrow 17 \rightarrow 21$ ladders with small γ steps), as the successful rescues of Section 3 demonstrate.

6 Repairing the double-double precision floor

The repository’s second solver family (a τ -ladder, spectral-style discretization in double-double arithmetic, ≈ 32 digits) had a long-standing self-residual floor of 2.7×10^{-17} where 10^{-29} should be attainable. A single-cell sensitivity probe (perturb one unknown by $h \in \{10^{-10}, \dots, 10^{-22}\}$, watch the operator response) localized the leak: the operator’s PCHIP interpolation *knots* were stored in float64 while everything else ran in double-double, capping the effective accuracy of the

interpolation map at knot precision. Making the knots double-double restored the expected floor:

$$F_{\text{pre}} = 4.31 \times 10^{-17} \quad \longrightarrow \quad F_{\text{post}} = 1.94 \times 10^{-28}.$$

Eleven digits recovered by a one-line-diagnosis fix — a reminder that in mixed-precision numerics the weakest stored object, not the weakest operation, sets the floor.

7 The strict- $h=0$ program: endogenous price-plane coordinates

7.1 The idea

All cube-based formulations share a defect: the level sets over which (1) integrates must be *re-derived* from sampled values, and any such derivation involves discrete decisions (which cells straddle the level) that make Φ discontinuous in the state. The resolution inverts the representation: **choose the price levels first, and make the grid points themselves the unknowns** — every grid point lives, by construction, on a plane of constant price, and the bandwidth h is identically zero with no limit taken. Three discretizations of this idea were built and stress-tested.

G1 -- Level surfaces $\{P^* = p\}$ for $p \in \{0.25, 0.4, 0.5, 0.6, 0.75\}$
Each surface carries its own endogenous grid (mesh vertices)

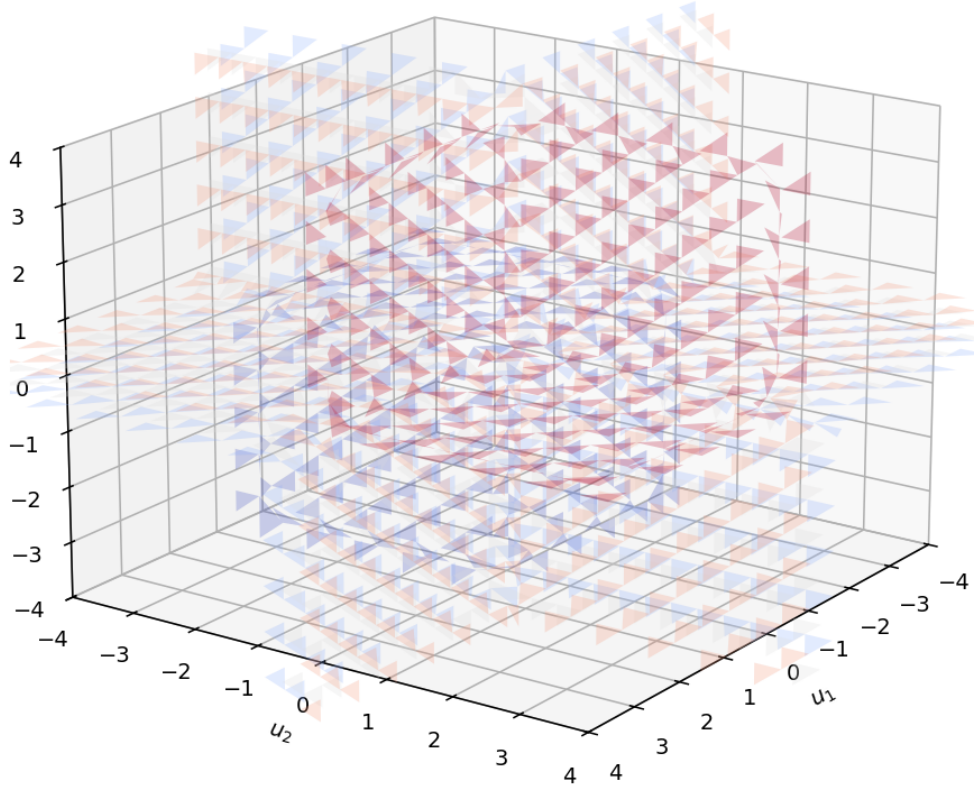
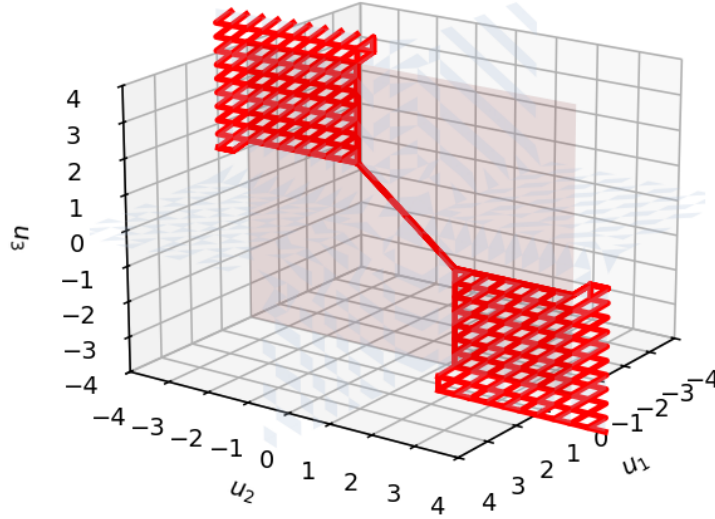


Figure 6: Level surfaces $\{P^* = p\}$ of the certified anchor equilibrium. In the endogenous-coordinate program these surfaces—not cube values—carry the unknowns.

7.2 Forward operator: validated at $h = 0$

Slicing a level surface by a trader’s own-signal hyperplane yields the 1-D evidence curve; arclength integration of $f_v f_v$ along it evaluates (1) exactly at $h = 0$ (Figure 7). The resulting forward operator was validated three ways: a closed-form pushforward unit test (empirical convergence order **2.007** in Δu); agreement with the kernel operator to $O(h^2)$ in the bulk and $O(h)$ near level-topology transitions (median 6×10^{-3} , tails 0.33 at $G=21$) — mapping exactly where the kernel’s bias lives; and a $9,000\times$ JIT speedup bringing a full-cube evaluation to 0.6 s.

G3 -- Hyperplane slice $\{u_1 = 0\}$ of the $p = 0.5$ surface
 Red curve = the strict- $h = 0$ integration domain



Same curve in the (u_2, u_3) plane -- a piecewise-linear polyline
 (black dots = segment endpoints from face crossings)

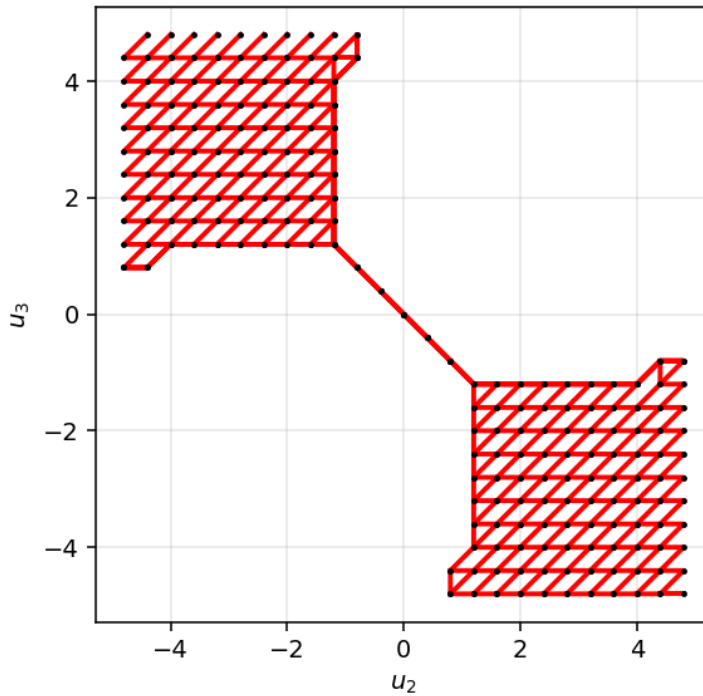


Figure 7: A price surface sliced by $\{u_1 = 0\}$: the red curve is the exact, bandwidth-free integration domain of the evidence integral.

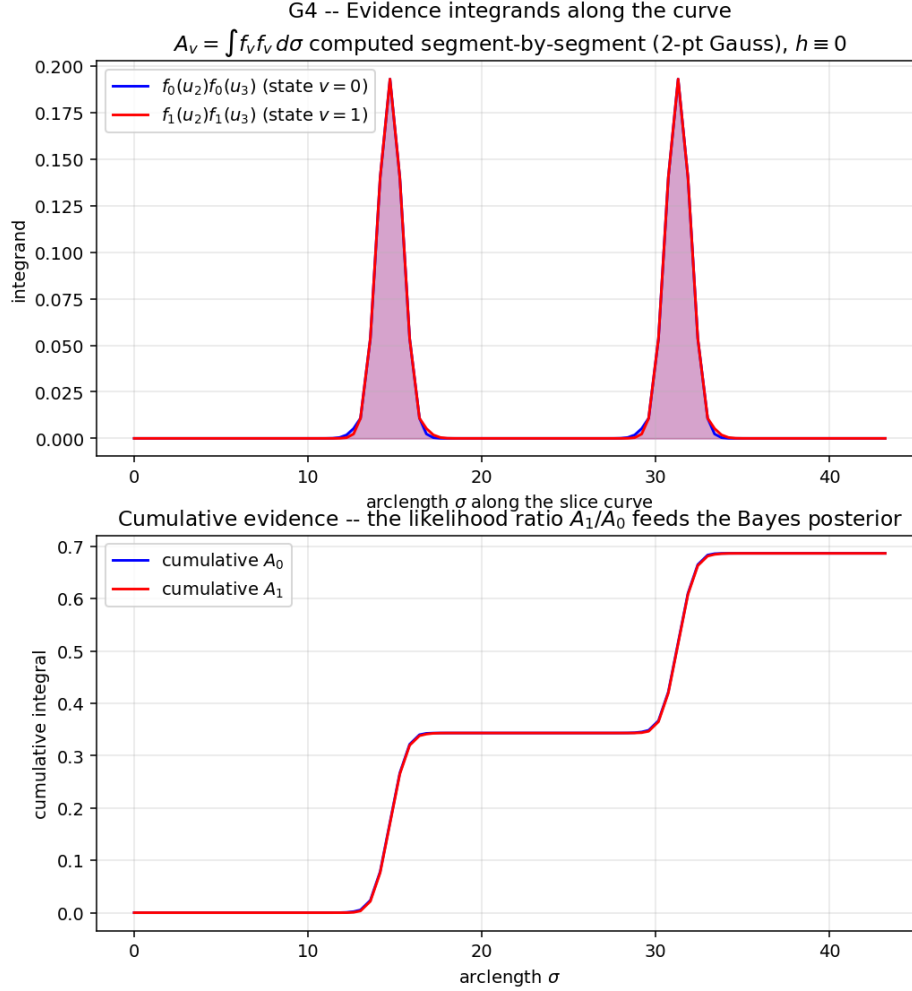


Figure 8: Evidence integrands and their cumulative integrals along a slice curve. The likelihood ratio A_1/A_0 feeds the Bayes posterior.

7.3 Two instructive failures and their lessons

Cube iteration diverges. Holding the representation on the cube and iterating with the strict operator diverges ($0.33 \rightarrow 0.46 \rightarrow 0.47$): marching-cubes re-extraction re-triangulates discontinuously as cube values move. *Lesson: the discontinuity is in the re-derivation, not the economics.*

Mesh LM plateaus. Representing each surface as a triangle mesh with *fixed topology* and free vertex positions (22,734 vertices; sparse-FD Jacobian exploiting the locality of vertex influence, validated exactly against dense FD, $11\times$ faster builds) produced the first genuinely descending strict- $h=0$ solver: 20+ accepted Levenberg steps, median residual $0.180 \rightarrow 0.130$. It then plateaued with damping $\lambda \rightarrow 10^5$: max residual pinned at 0.331 by a stubborn vertex set. Necessary safeguards discovered along the way: uniform (not Marquardt-scaled) damping, since the tangential gauge directions have near-zero curvature and Marquardt scaling under-damps exactly them; *per-vertex* step caps at a quarter of the smallest incident edge, since marching-cubes meshes contain slivers with 1000:1 edge ratios that any global step size flips; and a face-orientation topology guard. *Lesson: free-vertex motion with pinned topology is the right smoothness class, but unstructured marching-cubes meshes are a hostile numerical habitat.*

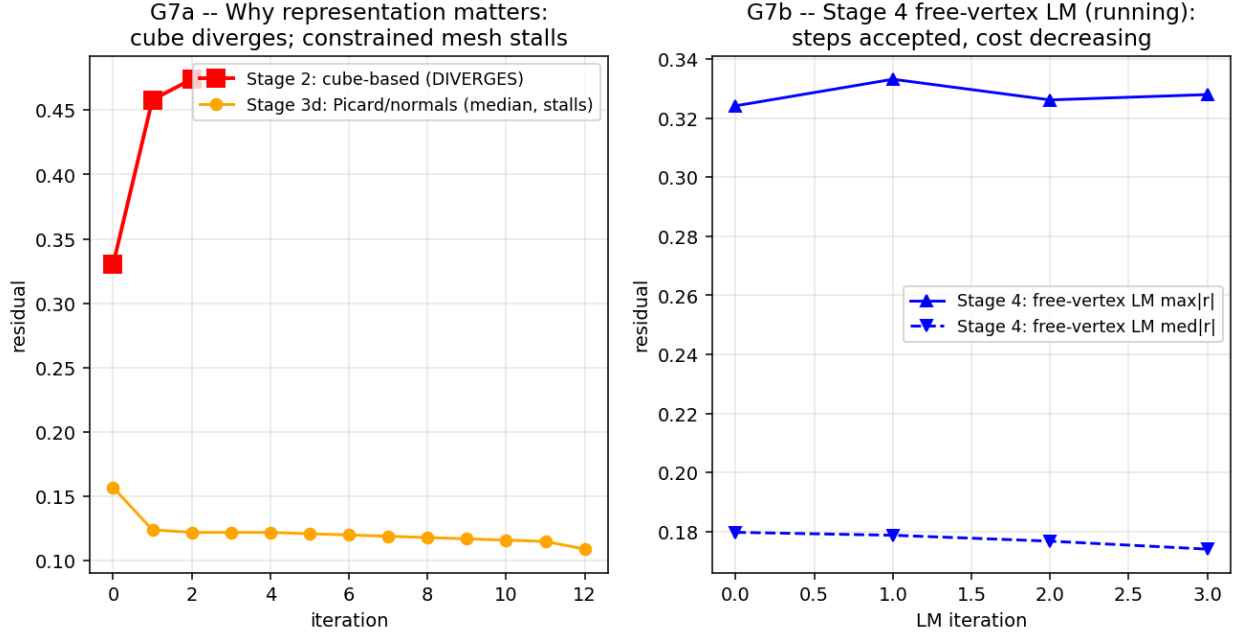


Figure 9: Left: cube-based iteration diverges; constrained (normal-only) motion stalls. Right: free-vertex LM descends steadily before its plateau.

7.4 The graph formulation

The final formulation exploits two structural properties to remove every remaining obstruction. *Monotonicity*: P increases in each signal, so each level surface is a graph $t = H_m(w)$ along the cube diagonal $t = (u_1 + u_2 + u_3)/\sqrt{3}$, and each slice curve is a graph over the anti-diagonal coordinate of the remaining two signals — level-set topology cannot change, ever. Slice curves are recovered from H_m by *monotone scalar root-solves*, smooth in the state by the implicit function theorem — a regular root, not a threshold decision. *Symmetry*: trader permutations act on the transverse plane as the dihedral group D_3 (storage on a $1/6$ wedge; one curve family serves all three agents), and the sign-flip symmetry $P(-u) = 1 - P(u)$ halves the surface count. The state shrinks $\sim 40\times$ relative to the mesh (no gauge freedom, no mesh management), and the Jacobian remains block-diagonal by surface.

7.5 The graph formulation’s verdict: strict $h=0$ is ill-posed on a dominance band

The graph solver passed every internal validation (root-solves exact to 9×10^{-14} ; initial residual statistics matching the mesh formulation, median 0.162 vs 0.180 — two independent discretizations agreeing on the strict- $h=0$ disequilibrium of the kernel solution), and on the well-posed part of the state it descends *far past* the mesh plateau: per-surface median residuals 0.0075–0.079 against the mesh’s 0.130 wall. It then exposed the true obstruction, which is mathematical, not numerical:

Tilt saturation. Slice-curve well-posedness requires the surface’s directional tilt to stay below $\sqrt{2}$ (the graph condition along the anti-diagonal). A tilt audit of the equilibrium surfaces shows 36–68% of the transverse domain at or beyond this bound on every surface, with mid-price surfaces saturating *exactly* at $\sqrt{2}$ through the high-density center. Tilt = $\sqrt{2}$ means the level surface locally

coincides with a plane $\{u_k = \text{const}\}$: the price is pinned by a *single dissenting trader's* signal. These are the CRRA one-dissenter clearing plateaus — visible as genuine atoms in the price distribution near $p = \frac{1}{3}$ and $p = \frac{2}{3}$ (two distinct quantile levels at 0.328 and 0.333).

Consequence. On the dominance band the conditioning event $\{P = p\}$ degenerates: the dissenter's evidence integral becomes square-root-singular in the surface state (finite-difference Jacobians blow up to $\|J\| \sim 10^6$) or exponentially insensitive (frozen residuals $|r| \sim 0.25$). Pseudo-arclength fold tracing, Tikhonov anchoring, and two-sided well-posedness filters all leave a plateau of $\max |r| \approx 0.27$ concentrated precisely on the band, and *mesh refinement makes it worse* (more singular rows at higher resolution): the plateau is intrinsic.

Interpretation. At exactly $h = 0$ the fixed-point equations are ill-posed on a positive-measure region of signal space whenever risk-averse clearing creates price atoms. The kernel bandwidth h is therefore not a numerical convenience but a *regularization* of a genuinely singular conditioning problem — traders conditioning on a price atom should condition on the *event* (positive probability mass), not on a measure-zero level set. Two principled routes follow: (i) h -continuation with Richardson extrapolation $h \rightarrow 0$ of observables (which the deep ladders of Section 4 already implement, explaining their success); or (ii) reformulating the equilibrium conditioning on the plateau event itself — an economically meaningful change of the information structure that we leave for future work.

8 Conclusions

1. **Certified low-residual equilibria now exist at scale:** 87 (τ, γ) cells with extended-precision residuals $\leq 10^{-15}$, quarantining everything else. The certified region's deficit map is the quantitatively trustworthy core of the project.
2. **The first continuum deficit estimates with honest error bars:** $\mathcal{D}_\infty = 0.268 \pm 0.010$ at the anchor, 0.2042 ± 0.0011 at $(\tau, \gamma) = (1, 0.098)$.
3. **The hard corner is solver-hard, not economics-hard:** spectra prove no fold through $\gamma = 1.2$ at $\tau = 2$; rescues recover formerly failing cells.
4. **Precision floors come from the weakest stored object:** the DD fix ($4 \times 10^{-17} \rightarrow 2 \times 10^{-28}$).
5. **Strict $h = 0$ is ill-posed where risk-averse clearing creates price atoms:** the endogenous-coordinate program delivered an exact, fast forward operator and successively eliminated every numerical obstruction (re-extraction discontinuity, mesh slivers, gauge freedom) until only a mathematical one remained — on the one-signal-dominance band the level-set conditioning degenerates, with the surfaces saturating the $\sqrt{2}$ tilt bound at the CRRA one-dissenter plateaus. The kernel bandwidth is the regularization of this singularity, vindicating the joint-limit $h \rightarrow 0$ extrapolation as the correct route to continuum values, and pointing to event-conditioning (atoms, not level sets) as the economically faithful reformulation.

Reproducibility

All data, scripts, figures, and per-cell solutions: `projects/REZN/solved_fixed_points/` subdirectories `dd.k3_overnight/` (`emin15/`, `deep_ladder/`, `stall_diagnosis/`, `anchor_sweep_20x5/`) and `cmm_endogenous/`; solver code in `projects/REZN/solver_code/highprec_dd.qd/` and `projects/REZN/solver_co`. The double-double knot fix is commit `2e0bd65e`.